THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

NORTH WESTERN ZONE FORM FOUR FORM PRE-NATIONAL EXAMINATION

TIME:3HRS **041 BASIC MATHEMATICS-17/9/2024**

INSTRUCTIONS

1. This paper consists of sections A and section B with a total of 14 questions.
2. Attempt all questions in both sections.
3. Each question in section A carries 6 marks while each question in section B carries 10 marks.
4. Mathematical tables and non-programmable calculator may be used.

SECTION A (60 MARKS)

1. (a) i. Every 5th person in a queue at the theatre will receive a free matinee pass (an afternoon performance in a cinema). Every 9th person will receive a free evening pass. Which person will be the first to receive both passes?

   (b) Fill in the following table by using classification of real numbers

	Number	Type	Decimal form	Types of decimal
	$\frac{1}{5}$	Rational	0.2	Terminating
1	$\frac{5}{13}$			
2	$\sqrt{9}$			
3	$\sqrt{13}$			

2. (a)Seypalm raised a certain number to the exponent of 5 and got 16 807. Which number did he raise to theexponent of 5?
   (b) Solve: $\log 25 + 2\log x = 2$

3. (a) In a group of 29 tourists from different countries, 17 went to Manyara National Park, 13 went to Mikumi National Park, and 8 went to neither Mikumi nor Manyara National parks. How many tourists went to both National Parks (Use only formula)

   (b) A family consists of 3 children of different ages. Find the probability of the following

   i. the oldest is a boy
   ii. the oldest and the youngest are boys

4. (a) Find the value of x and the equation of a line passing through (3, 8) and $(t, 5)$ parallel to another line whose equation is $6x - 4y - 3 = 0$

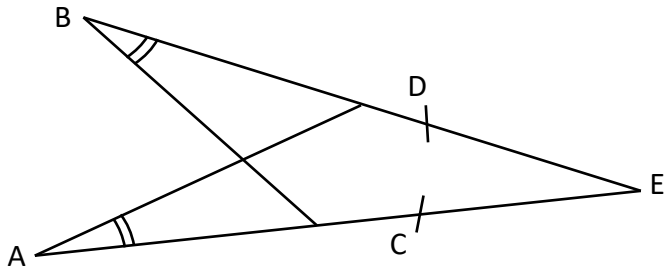
   (b) Given that vectors $\underline{c} = 3\underline{i} - 5\underline{j}$; $\underline{a} = 6\underline{i} - 2\underline{j}$; $\underline{c} = 2\underline{i} + 5\underline{j}$;. Find the following:

   i. $2\underline{a} + \underline{b} - \underline{3c}$

   ii. Scalar t and k if $t\underline{a} + k\underline{b} = 4\underline{i} - 3\underline{j}$

5. (a) The exterior angles of a three sided polygon are given in the ratio of 10: 9: 5. Find the interior angles

   (b) Given: $m\angle EAD \cong m\angle EBC$ and $\overline{BE} \cong \overline{AE}$. Prove: $\Delta BCE \cong \Delta ADE$



6. (a) y is directly as a cube root of x and $x = 2$ when $y = 9$
   i. Find the expression connecting the variables
   ii. Find y_2 when $x_2 = 8$

   (b) 40 men working 8 hours a day, finish a certain job in 16 days. Find how long would take the same job for 16 men working 10 hours a day? (Individuals are working at the same pace)

7. (a) Find the buying price of five text books sold at a constant price if one text book is sold at sh. 34 500 at a profit of 15%

   (b) Given the following balances, use them to prepare a profit and loss account for the year ended 31st December 2019.

Gross profit…… ….sh. 900 000
Wages and salaries……...sh. 100 000
Rent……sh. 50 000
Insurance…….sh. 120 000
Discount received…… ……sh. 80 000
Discount allowed…… sh. 300 000
Transport……. sh. 130 000
Commission received……sh. 400 000
Office expenses…….sh. 200 000
Bad debts…….sh. 20 000
Advertising……sh. 70 000
Sundry expenses……sh. 150 000

8. (a) Find the sum of the following series: $73 + 78 + 83 + \cdots 588$

   (b) Calculate $x\ and\ G_{15}$(the 15th term) so that $x - 1, x\ and\ x + 2$ are consecutive terms of geometric sequence.

9. (a) i. Which law either the law of cosine or the law of sine is used to calculate a? Give mathematical reason for the choice

   ii. Find a if $c = 8\ cm, b = 10\ cm, and\ m\angle A = 60^{\circ}$

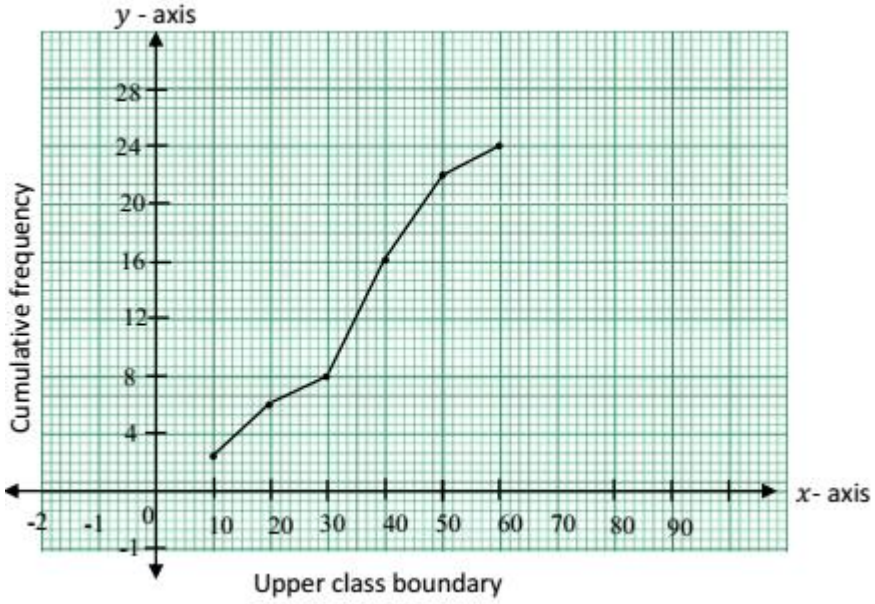
   (b) Find the third side of a right angled triangle whose two sides are $25\ cm\ and\ 20\ cm$

10. (a) Solve: $\begin{cases} \frac{m}{2} - \frac{n}{5} = 1 \\ 3n = 24 + m \end{cases}$

    (b) Solve by completing the square: $\frac{1}{x} + \frac{1}{x+1} = \frac{9}{20}$

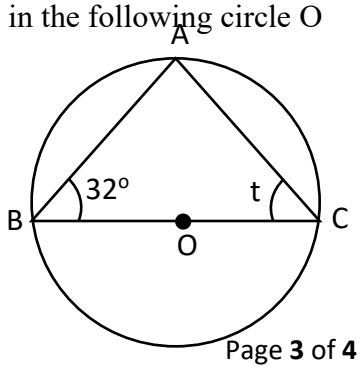
SECTION B (40 MARKS)

11. (a) Consider the following cumulative frequency curve as follows:



   i. Calculate the median score by using the formula
   ii. Use Histogram to estimate the mode score

   (b) Calculate angle t in the following circle O



12. (a) VABCD is a pyramid with $VA = VB = VC = VD = 5\ cm$ and ABCD is a square base of sides 4 cm each. Assume the centre of the base is at point N. Find the following:
    i. The angle between VA and the base ABCD
    ii. The volume of the pyramid

(b) How long will it take for a plane travelling at a speed of 650 km/h to travel between points Q$(30^{\circ}N,\ 28^{\circ}E)$and R$(30^{\circ}N, 72^{\circ}W)$?.

(Use $\pi = 3.142$ and the radius of the Earth R= 6 370 km)

13. (a) Find the image of the point (4,2) under a reflection in the line $-x + y = 0$ followed by a rotation through 90° clockwise direction about the origin.

(b) i. Calculate $x\ an\ d\ y$ so that

$$\begin{pmatrix} 2x & 4 \\ -3 & 5x \end{pmatrix} + \begin{pmatrix} 3y & -2 \\ -2 & -y \end{pmatrix} = \begin{pmatrix} -5 & 2 \\ -5 & 13 \end{pmatrix}$$

ii. Solve the following system of simultaneous equations by using the matrix method

$$5x - y = 7x + y = 12$$

14. (a) A furniture company has two plants that produce the lumber used in manufacturing tables and chairs. In 1 day of operation plant A can produce the lumber required to manufacture 20 tables and 60 chairs, and plant B can produce the lumber required to manufacture 25 tables and 50 chairs. The company needs enough lumber to manufacture at least 200 tables and 500 chairs. If it cost$1 000 to operate plant A for 1 day and $900 to operate plant B for 1 day
    i. How many days should each plant be operated to produce a sufficient amount of lumber at a minimum cost?
    ii. What is the minimum cost?

(b) Given: $(x) = -|x - 3| + 1$. Sketch the graph of the function and state the domain and range of f